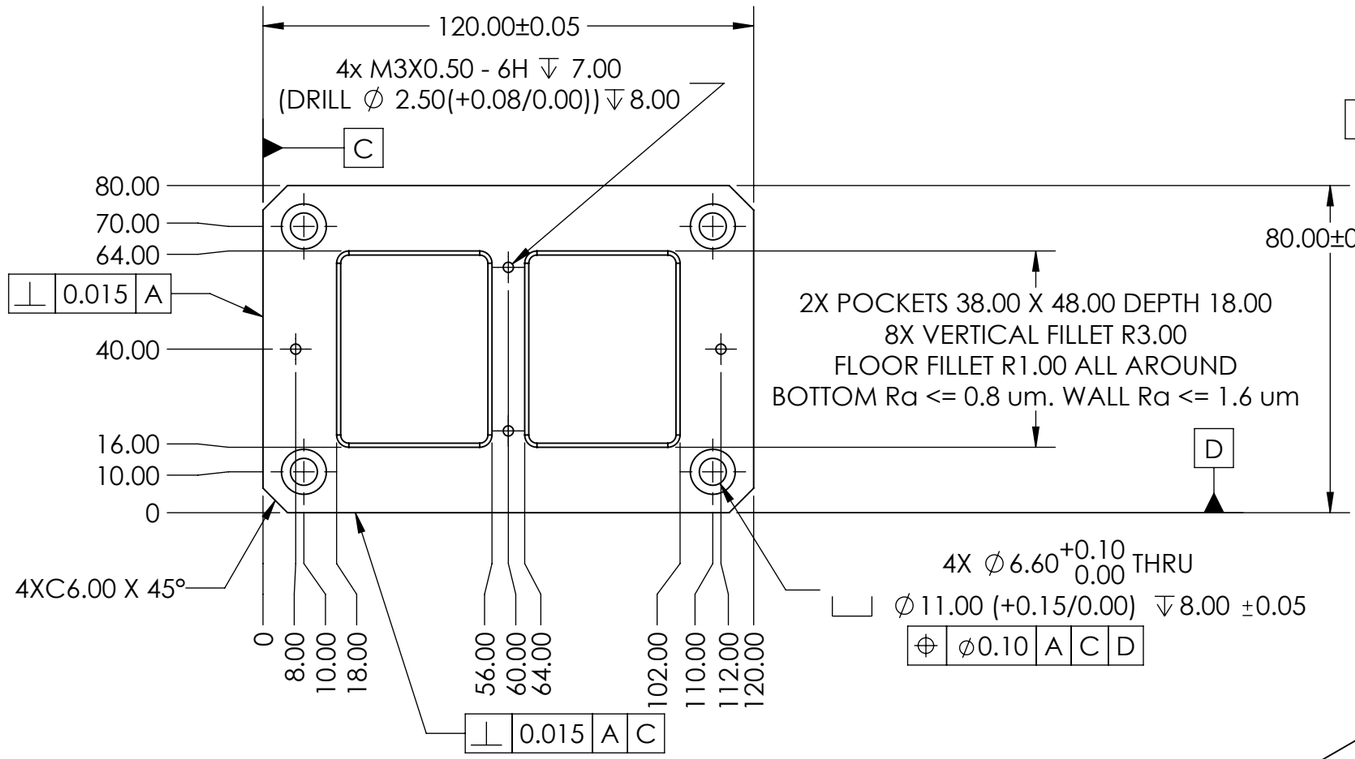
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

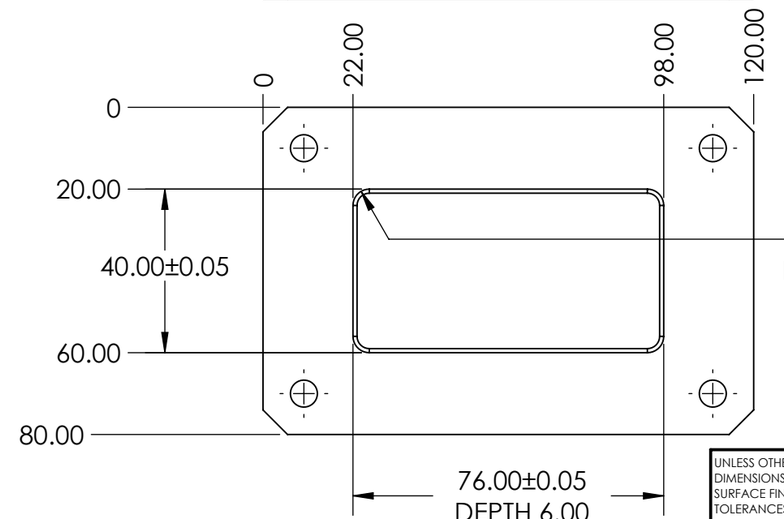
1. GENERAL & STANDARDS COMPLIANCE:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W3-EV-OBC-ENCLOSURE-4.4.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
   1.3 GENERAL TOLERANCES: LINEAR +/-0.05 mm, ANGULAR +/-0.2 DEG (5-AXIS PRECISION CNC MILLING).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
   1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

2. MATERIAL & EV AUTOMOTIVE RELIABILITY:
   2.1 MATERIAL OPTION A: FORGED AEROSPACE/EV ALUMINUM ALLOY 6082-T6 (EN AW-6082 / ASTM B221), THERMAL CONDUCTIVITY k >= 170 W/(m·K).
   2.2 MATERIAL OPTION B: FORGED ALUMINUM ALLOY 6061-T6 (AMS 4027).
   2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO DIE-CAST POROSITY, BRAZING FLUX, OR ADDITIVE RESIDUALS PERMITTED (IATF 16949 COMPLIANT).

3. THERMAL, EMI SHIELDING & GD&T CONTROLS:
   3.1 PRIMARY DATUM [A] (CHASSIS/COLD PLATE MATING FACE): FLATNESS <= 0.015 mm, SURFACE FINISH Ra <= 0.8 MICRONS.
   3.2 TOP MAGNETIC POCKET FLOORS [B]: PARALLEL TO [A] WITHIN 0.015 mm, Ra <= 0.8 MICRONS TO MINIMIZE THERMAL INTERFACE MATERIAL (TIM) CONTACT RESISTANCE.
   3.3 SOLID SHIELDING WALL (8.00 mm THICK): FULL SOLID SECTION TO PREVENT HIGH-FREQUENCY (150-300 kHz) MAGNETIC COUPLING AND HARMONIC CROSSTALK BETWEEN TRANSFORMER AND RESONANT INDUCTOR.
   3.4 4X CORNER MOUNTING COUNTERBORES (Ø11.00 mm): CENTERED AT (10.0, 10.0) CONVENTION TO ENSURE MINIMUM SURROUNDING SOLID WALL THICKNESS >= 4.40 mm.

4. LIQUID COOLING INTEGRITY & SURFACE FINISH:
   4.1 BOTTOM COOLING POCKET: DEPTH 6.00 mm, 4X VERTICAL CORNER FILLET R4.00 mm, FLOOR PERIMETER FILLET R1.00 mm TO MITIGATE CAVITATION RISKS AND FLOW DRAG.
   4.2 COOLANT PASSAGE PROOF PRESSURE: WITHSTAND 0.5 MPa (5.0 BAR) HYDROSTATIC PROOF TEST FOR 5 MINUTES WITHOUT LEAKAGE.
   4.3 CORROSION MITIGATION: ANODIZE PER MIL-A-8625 TYPE III HARD COAT (20-25 MICRONS THICK) ACROSS COOLING JACKET AND MATING FACES.





FOR REFERENCE

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm, Angular ±0.2° (5-Axis Precision CNC Milling)		FINISH: Hard Anodize Type III (MIL-A-8625, 20-25 µm thick, Anti-Cavitation & Anti-Glycol)		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE			TITLE: EV ON-BOARD CHARGER MAGNETIC INTEGRATED LIQUID-COOLED ENCLOSURE	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: Option A: Forged Aluminum Alloy 6082-T6 (EN AW-6082 / ASTM B221) / Option B: Al 6061-T6 (AMS 4027)		DWG NO. W3-EV-OBC-ENCLOSURE-4.4	A3
				WEIGHT: ~529.8 g (Al 6082-T6)		SCALE NTS	SHEET 1 OF 1